



TECHNOLOGICAL FUNDAMENTALS IN THE BIG DATA WORLD

CLUSTERING WITH KMEANS

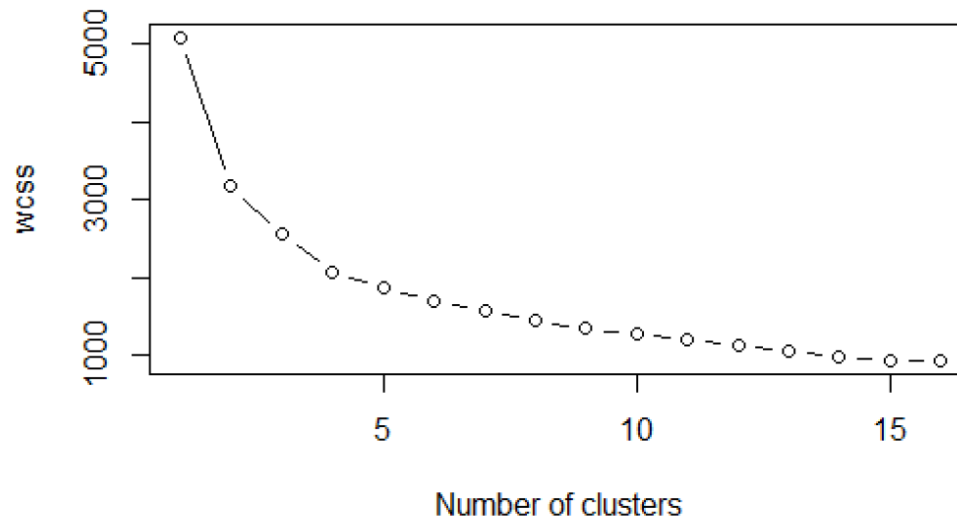
Lab 1

- To create a program to cluster data in a file with kmeans using Python.
 - ▣ File: proteins.csv

- You are asked to:
 - ▣ Write the serial version of the program in Python
 - ▣ Write the parallel version of the program in Python with multiprocessing and threads.
 - ▣ Write a memory explaining your results (maximum 12 pages)

- Notice:
 - ▣ You have fields in the dataset that are not numerical:
 - sequence
 - ▣ Change it by the sequence length when needed.

- Make a program in that:
 1. Constructs the elbow graph and find the optimal clusters number (k).
 2. Plots the results of the elbow graph. Choose optimum.



-
- Cluster plot
- Dim2 (15.6%)
- Dim1 (32.9%)
- cluster
- 1
 - 2
 - 3
 - 4
 - 5

5. For the cluster with highest sequence length, print id, highest value, average sequence length previously computed. If several, print taht with max. Hydrofob.
6. Prints a heat map with matplotlib.pyplot for the attributed of the clusters' centroids.

□ Multiprocessing version

1. Write a parallel version of you program using multiprocessing
2. Measure the time and optimize the program to get the fastest version you can.

□ Threaded version

1. Write a parallel version of you program using threads
2. Measure the time and optimize the program to get the fastest version you can.

- ❑ Make the lab in groups. Maximum 4
- ❑ Do not use Jupyter Notebook

- ❑ To deliver:
 1. authors.txt file
 2. Programs in Python: serial and parallel versions
 3. Written report (PDF).

- ❑ **Deadline: October 6th 2025. 23:30 hours.**
- ❑ **Important: deliver .py files. Jupyter notebooks not admitted.**